Exercises Session 2

Universidad Icesi

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1. Regular languages and automata theory

Exercise 1 *Given the languages* $A = \{a, b, c\}$ *and* $B = \{c, d, e\}$*, apply the following language operations:* $(A \cup B^2)$, $(AB)^*$ *y* $(BA)^R$.

Exercise 2 *Show that* $(A \cup B)^R = A^R \cup B^R$.

Exercise 3 *Show that* $(A^R)^R = A$.

Exercise 4 *Show that* $(A^+)^R = (A^R)^+$.

Exercise 5 *Given the languages* $L_1 = \{\lambda, 0, 10, 11\}$ *and* $L_2 = \{\lambda, 1, 01, 11\}$ *over the alphabet* $\Sigma = \{0, 1\}$*, obtain:* $L_1 \cdot L_2$, $L_2 \cdot L_1$, $L_1 \cup L_2$. $L_1 \cap L_2$, $L_1 - L_2$, $L_2 - L_1$, L_1^* *y* L_2^*.

Exercise 6 *Given the following languages:* $A = \{ab, b, cb\}$ *and* $B = \{a, ba\}$ *obtain the languages that result from applying the following language operations:* $(A \cup B^2)$, $(B \cup A)^R$, (AB), $(A^2 \cap BA)$ *y* $(A^R - B)^2$.

Exercise 7 *Given the languages:* $L_1 = \{01, 11\}$ *y* $L_2 = \{011, 101, 11\}$ *obtain the languages that result from applying the following language operations:* $(L_1 \cup L_2)^R$, $(L_2 - L_1)^2$, $(L_1 - L_2)^+$, $(L_1 \cap L_2)^*$ *y* $L_1^R L_2$.

Exercise 8 *Find the regular expression for the language over* $\Sigma = \{0, 1, 2\}$ *of all strings that start with* 2 *and end with* 1.

Exercise 9 *Find the regular expression for the language over* $\Sigma = \{0, 1\}$ *of all strings that start with two consecutive ones.*

Exercise 10 *Find the regular expression for the language over* $\Sigma = \{0, 1\}$ *of all strings that have a number of zeros divisible by three.*

Exercise 11 *Find the regular expression for the language over* $\Sigma = \{0, 1\}$ *of all strings that have at least two consecutive ones.*

Exercise 12 *Find the regular expression for the language over* $\Sigma = \{0, 1\}$ *of all strings that have at most two zeros.*

Exercise 13 *Find the regular expression for the language over* $\Sigma = \{0, 1\}$ *of all strings that only have one occurrence of three consecutive zeros.*

Exercise 14 *Find the regular expression for the language over* $\Sigma = \{0, 1\}$ *of all strings that have length of five or less.*

Exercise 15 *Find the regular expression for the language over* $\Sigma = \{0, 1\}$ *of all strings that start or end in* 00 *or* 11.

Exercise 16 *Find the regular expression for the language over* $\Sigma = \{0, 1\}$ *of all strings that have at least one* 0 *and at least one* 1.

Exercise 17 *Find the regular expression for the language over* $\Sigma = \{0, 1\}$ *of all strings that have at most two consecutive ones.*

Exercise 18 *Find the regular expression for the language over* $\Sigma = \{0, 1\}$ *of all strings which length is* ≥ 4.